Assignment 3: Github

Harsha Helsing, Manya Gupta

2026-11-09

Assignment 3: Collaborating in Github

Introduction

This is the base R markdown file we can populate with our code via branching, then we can replace this section with the introduction and data description, as well as the planned plots/analyses.

Installing and Loading Packages

```
#Feel free to add any and all packages we might need
#install.packages("pastecs")
library(pastecs)
#install.packages("tidyverse")
library(tidyverse)
#install.packages("knitr")
library(knitr)
#install.packages("car")
library(car)
```

Data Preparation

```
#Reading in csv file and assigning to data frame object named "text_wide"
text_wide <- read.csv("TextMessages.csv")
class(text_wide$Group)
```

```
## [1] "integer"
```

```
text_wide$Group <-factor(text_wide$Group,levels=c(1,2))

#In case we need long format data-
text_long <- pivot_longer(
  text_wide,
  cols= c("Baseline","Six_months"),
  names_to = "Timepoint",
  values_to = "Text_count"
  )
class(text_long$Timepoint)
```

```
## [1] "character"
```

```
text_long$Timepoint<-factor(text_long$Timepoint)

#Moving participant number to the front
```

```
text_wide <- text_wide |> relocate("Participant")
text_long <- text_long |> relocate("Participant")
```

Visualization 1: Faceted Boxplot of text messages by Group and Time

```
#Code chunk for boxplot
```

Visualization 2: Faceted Bar Chart of text messages by Group and Time

```
#Code chunk for bar chart
```

Summary Statistics of Text Messages by Group and Time

```
#We are first grouping the data by the group number and time-point, then
#computing the summary statistics that we want such as the mean, median, SD,
#range, etc., using dplyr from tidyverse.

sum_stat<- text_long |>
  group_by(Group,Timepoint) |>
  summarise(
  N=n(),
  Mean=mean(Text_count),
  SD=sd(Text_count),
  Median=median(Text_count),
  Min=min(Text_count),
  Max=max(Text_count),
  IQR=IQR(Text_count),
  .groups = "drop_last"
)

#To output a clean table of the summary statistics, we are using the kable
#function.

kable(sum_stat,
  caption="Summary Statistics of Text Messages by Group and Time-point",
  digits=2,
  align=c("c","l","c","c","c","c","c","c","c")
)
```

Table 1: Summary Statistics of Text Messages by Group and Time-point

Group	Timepoint	N	Mean	SD	Median	Min	Max	IQR
1	Baseline	25	64.84	10.68	64	47	85	16
1	Six_months	25	52.96	16.33	58	9	78	11
2	Baseline	25	65.60	10.84	65	46	89	14
2	Six_months	25	61.84	9.41	62	46	79	8

```
#At baseline, the two groups had relatively similar mean numbers of text
#messages. Group 1 had a mean of 64.84 (SD = 10.68), while Group 2 had a
#slightly higher mean of 65.60 (SD = 10.84). The similar means and standard
```

```
#deviations suggest that the groups had comparable text messaging patterns at
#the start of the study.

#At six months, both groups experienced a decrease in the mean number of text
#messages. However, the magnitude of the decrease differed between groups. Group
#1 had a mean of 52.96 (SD = 16.33), representing an average decrease of 11.88
#messages from baseline. Group 2 had a smaller decrease, with a mean of 61.84
#(SD = 9.41), representing a reduction of 3.76 messages.

#When comparing means to medians, most of the subgroups have similar values
#hinting at symmetrical distributions. We only see a major difference in the six
#month time-point for group 1 where the median was higher than the mean,
#suggesting a left skew to the distribution. There were also differences in the
#variability of text messaging patterns at six months. Group 1 demonstrated
#greater variability (SD = 16.33) compared to baseline, suggesting that
#participants' messaging behaviors became more dispersed. This is also reflected
#in the wider range of values (9-78). In contrast, Group 2 demonstrated less
#variability at six months (SD = 9.41), with a narrower range (46-79),
#suggesting more consistent messaging patterns among participants.
```

Interpretation and conclusions

Any concluding comments in addition to individual interpretations above

Summary table conclusions

Overall, both groups demonstrated a reduction in text messaging frequency over the six-month period, but the decrease was more pronounced in Group 1. While the groups had comparable baseline averages, their six-month averages diverged, with Group 1 reporting fewer messages on average than Group 2.

These descriptive findings suggest a potentially greater change in messaging behavior in Group 1. However, descriptive statistics alone cannot establish whether the observed differences are statistically significant or attributable to group assignment. Further inferential analysis, such as comparing within-group changes and between-group differences in change, would be necessary to determine whether the differences represent meaningful intervention effects.

One statistical consideration: Because the same participants were measured at baseline and six months, a paired analysis would be appropriate for evaluating changes within each group.